

Akhilesh Anant Gonabal

ROBOTICS AND EMBEDDED SYSTEMS ENGINEER

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Education

National Institute of Technology Karnataka

B.TECH IN ELECTRICAL AND ELECTRONICS ENGINEERING

Surathkal, India

Aug 2016 - June 2020

Key Coursework: Digital & Analog Systems, Microcontrollers & Embedded Systems, Control Systems, Sensors & Actuators, Machine Learning

Work Experience

Software Engineer Level 2 | CAPLA India Pvt Ltd, Bengaluru, India

Sept 2025 - Present

- Build and refine C++/Qt/QML features for Linux-based embedded medical-technology software on Jetson-class platforms, focusing on hardware-connected workflows, responsive UI behavior, and operator-ready tooling.
- Debug and validate imaging, tracking, and communication paths through structured testing and issue triage, turning robotics, computer vision, and embedded systems expertise into robust hardware-integrated workflows.

Research Assistant | Indian Institute of Science, Bengaluru, India

May 2024 - May 2025

- Developed a multimodal robotic system for constrained-space inspection, integrating Intel RealSense depth sensing, Jetson-class compute, embedded controllers, and ROS 2 teleoperation workflows.
- Designed distributed embedded and mechatronic subsystems, including modular wall-press mechanisms, 2-DoF joints, and CAN-based coordination for pipe traversal and field-oriented validation.

Junior Research Fellow | National Institute of Technology Karnataka, Surathkal, India

Sept 2022 - May 2024

- Built an Applied Cyber-Physical Systems lab platform using STM32/ESP32 sensor-actuator nodes, edge-fog-cloud architecture, MQTT, and ThingsBoard dashboards for hands-on education and research.
- Developed remote experimentation workflows for 40+ students with real-time analytics and web-based access, improving practical understanding by 28% as measured by course feedback.

Lead Engineer | TATA Power Company Ltd, Mumbai, India

Sept 2020 - Jan 2022

- Developed a SCADA-integrated operations and maintenance (O&M) dashboard with automated email/SMS alerts for critical substation faults, reducing customer power downtime by 30–40% through proactive monitoring.
- Automated SCADA system health monitoring and periodic reporting by analyzing three years of historical data, improving operational efficiency by 60–70% and enabling data-driven maintenance.

Patents and Publications

Patent: “A System for Inspection and Maintenance of Complex Environments using Multiple-Modal Operations and Method Thereof”

Dec 2024

GRANTED | PATENT NO. 591148

Publication: “Lab Experimental Framework for Demonstrating Integral Concepts of Applied Cyber-Physical Systems”

Dec 2023

11TH ISSS NATIONAL CONFERENCE ON MEMS, SMART MATERIALS, STRUCTURES AND SYSTEMS

Projects

SMART CITY

Feb 2019 - Apr 2019

PROF. GANGADHARAN K. V | PYTHON, C, ARDUINO, RASPBERRY PI, ESP8266, MQTT, JAVASCRIPT, HTML

- Engineered a scalable IoT-driven smart city test-bed, integrating diverse sensors, actuators, and an MQTT-enabled communication network to demonstrate autonomous resource management and system-wide interconnectivity.
- Implemented a centralized command system for real-time monitoring and control of all smart city modules, coordinating autonomous tasks through sensor data and decision logic.

HOMEMAKER ASSISTANCE SYSTEM

Aug 2018 - Apr 2019

PROF. JORA M GONDA | RASPBERRY PI, ESP8266, C++, PYTHON, DJANGO, HTML, JAVASCRIPT, CSS

- Developed a Raspberry Pi-based voice-activated home automation system tailored for disabled and elderly users, integrating ESP8266 and GSM modules with sensors and actuators to enable hands-free device control and improve accessibility.
- Implemented real-time alerts and notifications for critical home events, providing immediate feedback to users, and enhanced system usability through iterative testing to meet diverse accessibility needs.

Skills

Programming Languages C/C++, Python, MATLAB, JavaScript, HTML/CSS, LaTeX

Hardware & Embedded Platforms Jetson Nano, Raspberry Pi, STM32, Teensy, Arduino, ESP32, ESP8266, Intel RealSense

Frameworks & Utilities ROS 2, Gazebo, FreeRTOS, MQTT, CAN, MQTT Explorer, Postman, Wireshark, Docker, Git, CMake, ThingsBoard

Software & Simulation Tools Qt/QML, KiCad, Altium, Onshape, NI LabVIEW, SolidWorks, Fusion 360, MATLAB, Simulink

Operating Systems Linux, Microsoft Windows